CLASS 9 — MATHS · FORMULA SHEET

Chapter 4: **Algebraic Identities** | Quick revision, one page

Square identities

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a+b)(a-b)$
- $(x+a)(x+b) = x^2 + (a+b)x + ab$

Cube identities

- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- Sign pattern in $(a-b)^3$: the signs alternate $+, -, +, -$.

Useful consequences

- $a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$
- $(a+b)^2 - (a-b)^2 = 4ab$ and $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$
- Numerical use: $103^2 = (100+3)^2$, $98^2 = (100-2)^2$, $102 \times 98 = 100^2 - 2^2$, $11^3 = (10+1)^3$.

Factorisation

- Take out the common factor first, then look for an identity.
- $x^2 + (a+b)x + ab = (x+a)(x+b)$: find two numbers with sum $= a+b$ and product $= ab$.
- $4x^2 + 12x + 9 = (2x+3)^2$ (perfect square: first and last terms are squares, middle term is twice their product).
- A difference of two squares factorises as $(a+b)(a-b)$; a sum of two squares does not factorise over real numbers.

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